Detecting Outliers in Multiclass Stellar Populations via Isolation Forest Anomaly Detection Frameworks

Shriya Kunnanath, Naailah Sheik, Jiyaa Solanki, Safdar Zaman, Isabelle Niu, Michael Shohat

ABSTRACT

Leveraging high-precision stellar measurements, this study explores anomaly detection in large-scale astrophysical datasets. Querying, cleaning, and processing data from the Gaia DR3 catalog, the stellar sample was categorized according to the Morgan–Keenan classification system to provide a structured framework for analysis. To identify anomalies, we applied an Isolation Forest anomaly detection algorithm with varying hyperparameters, running multiple iterations to assess the consistency of outlier detection. Silhouette Scores were then calculated to evaluate the clustering quality of each contamination level, helping to fine-tune the model before further stellar classification. Our results revealed that lower contamination values achieved higher silhouette scores, with the best performance reached at a contamination parameter of 0.005 and n_estimators of 500, producing a silhouette score of 0.6191. These findings demonstrate Isolation Forest's effectiveness in detecting stellar anomalies in Gaia-scale datasets and highlights its potential to uncover rare stellar populations that may advance our understanding of stellar evolution and classification.

INTRODUCTION:

The rapid growth of stellar datasets reveals an immense number of stars, each with different characteristics such as temperature, luminosity, color, and parallax—the difference in direction between two different points. The Hertzsprung-Russell (H-R) Diagram is a diagram that classifies stars based on luminosity and temperature. Naturally, within these vast datasets lie anomalies—a star that doesn't follow the trends seen in the H-R Diagram. Anomalies are the

key to discovering new aspects of our universe as they provide insight into the formation, dynamics, and processes of the cosmos. Identifying the true number of outliers is challenging due to the complexity and magnitude of data (Liu et al.). Manually using statistics, classification, and clustering isn't an efficient way to highlight anomalies within datasets due to time consumption and the inability to identify subtle anomalies. They analyze normal instances of stellar clustering instead of anomalies. Thus, with large amounts of data, it is hard to detect the true number of stellar anomalies.

Isolation Forest (IF) has been widely used in anomaly detection due to its efficiency, scalability, and ability to handle high-dimensional datasets without requiring extensive parameter turning. IF is an unsupervised machine learning model, a type of algorithm that identifies patterns from unlabeled data and creates classes based on those patterns. Designed to detect anomalies, IF breaks the data into trees (iTrees) and sections them based on differences within different characteristics measured in the dataset, otherwise known as features. This creates partitions for data, and outliers can be identified if each point can't split further. When performing random partitioning, iTrees result in shorter paths and a less complex structure for specific points, hence identifying anomalies (Liu et al.). Their research signified that IF can effectively distinguish anomalies from regular data points while maintaining more than a linear time complexity. Its design allows it to detect anomalies during high-dimensional problems with a short run time. Furthermore, IF continues to be coherent and efficient even when the given data set contains zero anomalies. Zhuliang Qi utilized a random forest algorithm for the classification of celestial bodies, which resulted in a 98% accuracy rate. The decision tree model also used in the research displayed a 97% accuracy rate, yet demonstrated the greatest behavior in stellar classification. This process makes it more effective than other methods for handling large datasets. Identifying stellar anomalies in vast datasets similar to Gaia is more efficient and

reliable when using Isolation Forest because of its ability to process and classify large sets of data.

This paper examines the accuracy and productivity of Isolation Forest's ability to detect stellar anomalies, along with the trends and phenomena that can be identified.

DATA ACQUISITION:

The astronomical data in this study was obtained from the Gaia mission, a space observatory launched by the European Space Agency (ESA). Specifically, this study incorporated a subset of the Gaia Data Release 3 (DR3) catalog, which provides accurate astrometric, photometric, and spectrometric data for over a billion stars in the Milky Way. This data was sourced from publicly available datasets, and was cleaned to remove irrelevant anomalies, forming the basis for the stellar anomaly classification.

In the dataset, each element represented a single star, identified by a unique numerical catalog ID. For each star, parallax, parallax_error, abs_g_mag, phot_g_mean_mag, phot_bp_mean_mag, and bp_rp were the features queried. These features provided essential information about the star's position, brightness, color, and temperature. The parallax and parallax_error values made it possible to estimate the star's distance and assess the uncertainty in this measurement. The phot_g_mean_mag and phot_bp_mean_mag magnitudes are key to calculating the star's absolute magnitude (abs_g_mag), which reflects its intrinsic brightness. This value is essential for determining the star's luminosity and plays a central role in classifying its type, whether as a main sequence star, giant, or white dwarf. The bp_rp color index, which measures the difference between the blue and red photometric bands, was used to estimate the star's effective temperature (Teff). These Teff values, in turn, allowed for the classification of stars into spectral types, ranging from the hot O-type to the cooler M-type stars,

based on established temperature thresholds. These features–absolute magnitude, color index, and effective temperature–formed the foundation for the stellar classification and the identification of potential anomalies within the dataset.

Initially, the Gaia query acquired around 500,000 elements of stars, providing a sufficiently large sample size to ensure coverage for detecting stellar anomalies. However, to preserve the reliability and scientific value of the dataset, a data-cleaning process was applied to retain only high-quality and astrophysically meaningful stellar entries. Specifically, rows with missing or invalid values in key columns like parallax, phot_g_mean_mag, and bp_rp, and excluded non-positive parallax values were removed. Stars with parallax errors greater than 20%, ensuring phot_g_mean_mag remained between 0 and 25 were also filtered. Following the series of data-cleaning steps, which removed most of the unusable elements, the final dataset ended up consisting of around 98,500 stars, representing 19.7% of the originally retrieved sample.

Following the Morgan Keenan (MK) classification system, which assigns stars both a luminosity class and spectral class, the cleaned data was then categorized into its respective classes. The luminosity classes, determined by the absolute magnitude value, consisted of main sequence stars, giants, white dwarfs, and subgiants. On the other hand, the spectral classes, determined by effective temperature, ranged from O, B, A, F, G, K, to M, with class O being the hottest and class M being the coolest. The MK classification system utilized calculated features, which included distance, log_L, and Teff. Log_L was used to estimate the points' luminosity while the Teff value was used to categorize the points into their spectral type. In order to calculate Log_L, this study also determined distance by finding the inverse of the points' parallax values and converting them to parsecs.

To visualize the classifications, the cleaned and categorized data was then plotted on an HR diagram. The HR diagram categorized data points into its respective luminosity and spectral classes based on their surface temperature (in degrees Kelvin) and their absolute magnitude, with the luminosity classes displayed as different colors.

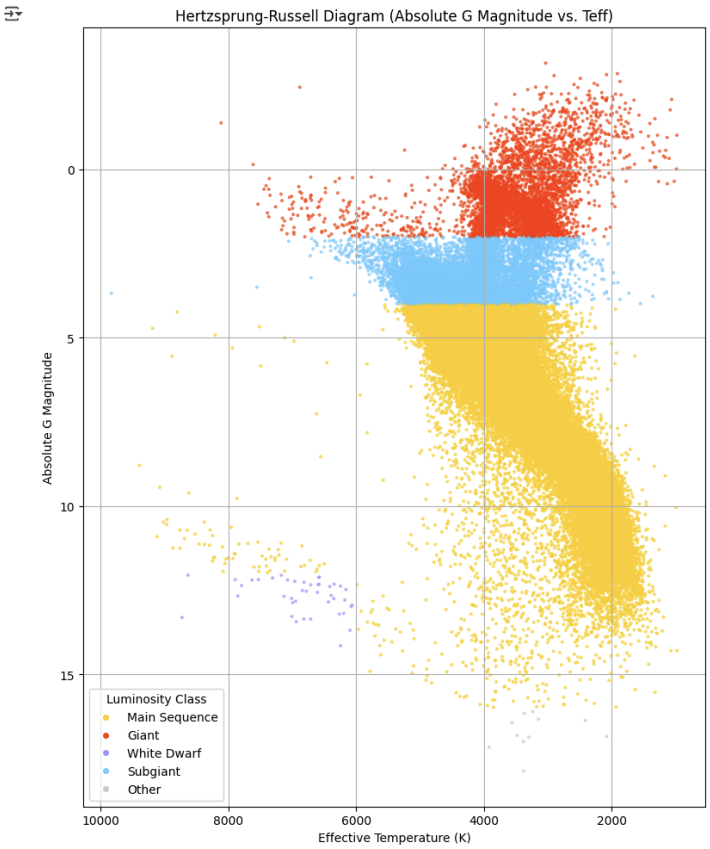


Figure 1: HR diagram of the study's dataset

As demonstrated in the figure above, the HR diagram depicted an excessive amount of the dataset being classified into main sequence stars with temperatures under 5000 K, and a significant amount of giants and supergiants. This visualization highlights how white dwarfs and stars in the lower left portion of the diagram are significantly lower in number and therefore part of the minority class. To elaborate on this data imbalance, the frequencies of each luminosity class and spectral class were plotted. Evident in these illustrations, it was confirmed

that the other and white dwarf stars and spectral types A and F were infrequent. This data imbalance indicates that the IF model will likely classify elements within the minority class as anomalies.

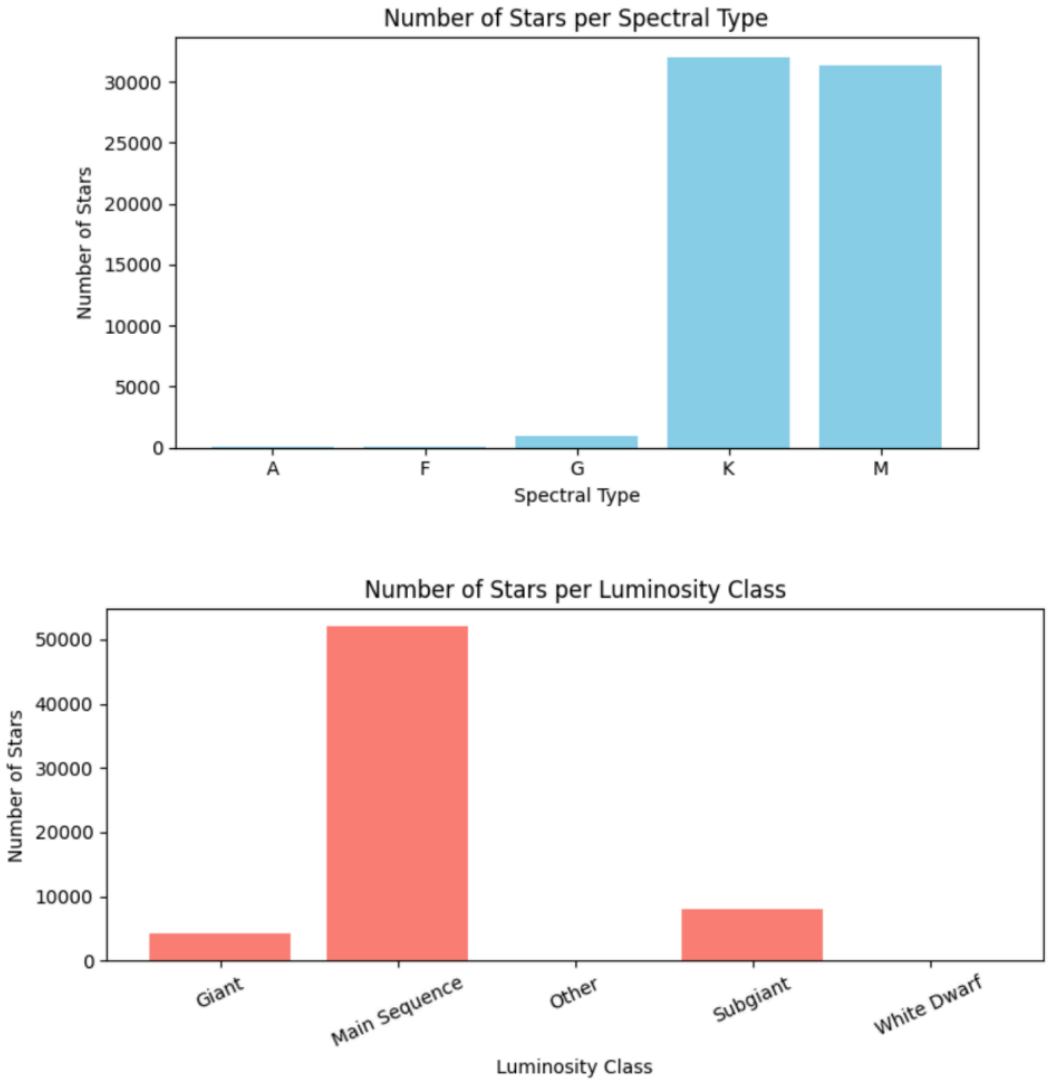


Figure 2: Bar charts of element frequencies of spectral type and luminosity classes in the cleaned and processed dataset.

The template contained herein is provided "as is." UpCounsel, Inc. does not provide any express or implied warranties of merchantability, suitability, or completeness of the information in this template. You use this template at your own risk. Neither UpCounsel, Inc., nor its agents, officers, employees, or affiliates, are liable for any direct, indirect, incidental, special, exemplary, or consequential damages (including procurement of substitute goods or services, loss of use or profits, or business interruption), even if UpCounsel, Inc. has been advised of the possibility of such damages, on any theory of liability, whether in contract, strict liability, or tort, arising in any way out of the use of or inability to use this template. Some jurisdictions do not allow the limitation of incidental or consequential damages, so this limitation may not apply to you.

MODEL CREATION:

This study utilized the IF model algorithm to detect anomalous stars in the Gaia dataset. During the study, two key hyperparameters were altered: n_estimators and contamination. N_estimators outlines the number of isolation trees in the isolation forest algorithm. Higher values of n_estimators lead to more stable anomaly detection as the model takes the results from multiple trees to understand the difference between anomalies and normal data points. The contamination hyperparameter is the expected ratio of outliers in a dataset; it acts as a determining factor for the number of anomalies a dataset should have. When there is a lower contaminator hyperparameter value, it indicates that the Isolation Forest model only labels obvious points as outliers. If there is a higher contamination hyperparameter, the model will be more lenient and label more points as anomalies. Additionally, the IF model also relied on random state and sample size. Keeping the random state at 42 ensures that the results are reproducible and consistent across multiple runs. Without this, the model could vary with reproducibility. The sample size ensures that the dataset is capped at a certain number of data

points used to train each isolation tree, which helps produce the same results every time our algorithm runs.

The IF model was trained on stellar data with three physical features: effective temperature (teff), logarithmic luminosity (log_L), and the bp-rp color index (bp_rp). These features were selected as they showed the strongest significance during the classification of Morgan-Keenan classes, indicating that they play an important role in determining anomalies. Instead of utilizing a broader set of ten or more features, three were utilized to detect anomalies efficiently. Inputting the IF model with too many stellar features leads to overfitting and reduced model interpretability. Using an excessive number of input features could result in the curse of dimensionality, where increasing dimensions can lead to growth in space volume, which makes data points spaced and hard to analyze.

The IF model produced two outputs for each start: anomaly_score and anomaly. Anomaly score is a value that represents how isolated a data point is. A lower value will correspond to points that are more likely to be anomalies. Anomaly was a binary classification where -1 would signify an outlier, while +1 would indicate an inlier.

Once the IF model had been initialised and trained we used Silhouette scores to test the accuracy of the model. Silhouette score is a suitable option to test the model's accuracy as it identifies data points with high or low scores depending on how well-defined and accurately identified the anomalous data points are. Silhouette scores are calculated by the following equation.

$$s(i) = \frac{b(i) - a(i)}{\max\left(a(i), b(i)\right)}$$

Figure 3: Equation to derive Silhouette Score.

a(i) is the average distance from point *(i)* to the other points on the cluster and *b(i)* is the minimum average distance from point *(i)* to all points in any other cluster.

These scores are a number which lies between '1' and '-1' inclusive. Negative scores indicate low accuracy meaning the anomalies detected are misclassified while positive scores indicate high accuracy verifying that the anomalies detected are accurate and well defined. Hence, obtaining a positive silhouette score closer to '1' is aimed for. In order to test how the IF model's performance changed with different hyperparameter configurations silhouette scores were calculated first by varying hyperparameter values of "contamination" (0.005, 0.01, 0.05, 0.1) while keeping the hyperparameter "N_estimators" constant at the value of 500, then by varying hyperparameter values of "N_estimators" (100, 200, 500) while keeping the hyperparameter value of "contamination" constant at the value of 0.005. In each of these combinations we keep the hyperparameter "random_state" constant with a value of 42 to ensure reproducibility.

RESULTS:

Silhouette score calculations were used in this study to determine how successfully each data point was categorized into its respective classes. It measures how similar a data point is to

its own cluster and other clusters. The formula for a silhouette score is represented as [b(i) - a(i)]/[max(a(i),b(i))] where a is the average distance from point i to other points in the cluster and b is is the minimum average distance from point i to all points in any other cluster. A silhouette score can range from -1 (assigned to the wrong cluster) to 1 (assigned to the right cluster). A score from 0.5 to 1 would signify that the data point was classified into an appropriate cluster. Conversely, a score ranging from -0.5 to -1 would suggest that the data point is closer to points in other clusters instead of its own.

```
Average Silhouette Score for n_estimators=500, contamination=0.005: 0.6191
Average Silhouette Score for n_estimators=500, contamination=0.01: 0.5637
Average Silhouette Score for n_estimators=500, contamination=0.05: 0.3882
Average Silhouette Score for n_estimators=500, contamination=0.1: 0.3290

Top Average Silhouette Scores:
   n_estimators  contamination  average_silhouette_score
0           500          0.005                  0.619136
1           500          0.010                  0.563657
2           500          0.050                  0.388206
3           500          0.100                  0.328974
```

Figure 4: Silhouette Scores for varied hyperparameter configurations.

As demonstrated above, the silhouette scores varied based on their contamination parameters. The greatest silhouette score was 0.6191 for the contamination parameter of 0.005, indicating that this parameter has the best clustering ability. Conversely, the contamination parameter of 0.1 had the lowest silhouette score of 0.3290, indicating that this parameter was worse at assigning data points to the appropriate cluster compared to the other contamination parameters.

Given that each of the n_estimators were equal, the results may suggest that a lower contamination parameter is ideal for producing a high silhouette score.

When compared to other studies, silhouette scores above 0.5 is a good indication of effective clustering. Some studies have achieved scores as high as 0.7 and 0.8 in cases where the data is well-separated and the clusters are distinct. A score between 0.3 and 0.5 suggests that the clusters are meaningful but may have some overlap or unclear boundaries. Scores below 0.3 generally indicate that the clustering algorithm has not effectively separated the data into meaningful groups. This also signifies that the algorithm used was possibly not suitable for the given data set. In Density-Based Spatial Clustering of Applications with Noise (DEBSCAN) studies, a contamination parameter between 0.01 and 0.05 usually yields the best results, as proved by the 0.05 parameter used in this study. Similarly, in K-means clustering, silhouette scores between 0.5 and 0.6 is when the data is well-separated and data points are closer to elements in their respective clusters. However, when there is a significant amount of overlap or when clusters are poorly chosen, the scores tend to drop closer to 0.3 and 0.4, much like the lower contamination results in this study. For other studies it's also common to see a negative correlation between contamination rates and silhouette scores, meaning that as contamination parameters increase, silhouette scores decrease. The results of this study follow that trend, where contamination values closer to 0.05 yield higher silhouette scores while larger contamination values closer to 0.1 result in lower scores.

CONCLUSION:

The IF model could be created to incorporate more input features such as "parallax" in addition to the existing input features "Teff" and "log_L". These features can also be varied in combinations by adding or subtracting a specific feature in order to test which combination of features produce the most accurate results.

The Gaia database has heteroscedastic errors which means that the measurement errors are not constant instead they depend on the individual objects. In the Gaia database bright stars have a low parallax error value while fainter or crowded-field stars have greater parallax error values. The issue arises when these raw parallax error values are input into the IF model as a faint star with a large parallax error might seem anomalous due to its values being larger instead of actually being astrophysically strange. To avoid this, the Monte Carlo feature sampling method may be used. The Monte Carlo feature sampling method builds a multivariate Gaussian distribution centered at its measured values while the spread and correlation come from Gaia's provided uncertainties and covariance matrix. Then multiple samples are taken consisting of many synthetic copies of the star's features from this distribution. Each sample is then input into the IF model to obtain an anomaly score. Finally an average of these scores is calculated to receive an anomaly score conscious of the varying parallax errors.

Stellar classification involves the clustering of different star classifications. Anomaly detection can be used to predict anomalous stars which means that they do not belong to any of the pre-existing star classifications. If enough anomalies are detected with similar astrophysical features they may become a new classification furthering our understanding on the different types of star. Anomaly detection may also be used to identify stars with rare features such as white dwarfs with different colour/luminosity features. After having identified these peculiar stellar anomalies they can be studied more closely to extrapolate why they have these rare characteristics.

The IF model was found to perform the most accurately on the hyperparameter settings of "contamination" value 0.005 and "n_estimators" value 500 seen from the silhouette scores obtained from various hyperparameter combinations as the highest silhouette score 0.6191 was

obtained with this combination. While the silhouette scores were seen to decrease as the "contamination" hyperparameter value increased and vice versa.

REFERENCES

Abdel-aziem, Ayman H., and Tamer H. Soliman. "A multi-layer perceptron (MLP) neural networks for stellar classification: A review of methods and results." *International Journal of Advances in Applied Computational Intelligence*, vol. 3, no. 2, 2023, pp. 29–37, https://doi.org/10.54216/ijaaci.030203.

Al Farizi, Wahid Salman, et al. "Isolation Forest based ANOMALY DETECTION: A systematic literature review." *2021 8th International Conference on Information Technology, Computer and Electrical Engineering (ICITACEE)*, 23 Sept. 2021, pp. 118–122, https://doi.org/10.1109/icitacee53184.2021.9617498.

Hariri, Sahand, et al. "Extended isolation forest." *IEEE Transactions on Knowledge and Data Engineering*, vol. 33, no. 4, 1 Apr. 2021, pp. 1479–1489, https://doi.org/10.1109/tkde.2019.2947676.

Liu, Fei Tony, et al. "Isolation Forest." *2008 Eighth IEEE International Conference on Data Mining*, Dec. 2008, pp. 413–422, https://doi.org/10.1109/icdm.2008.17.

Qi, Zhuliang. "Stellar Classification by machine learning." *SHS Web of Conferences*, vol. 144, 2022, p. 03006, https://doi.org/10.1051/shsconf/202214403006.

Shy, Sarah, et al. “Incorporating measurement error in astronomical object classification.” *The Astronomical Journal*, vol. 164, no. 1, 13 June 2022, p. 6, https://doi.org/10.3847/1538-3881/ac6e64.

Yang, X, et al. “Searching for short-timescale radio anomalies using nonlinear dimensionality reduction techniques.” *Monthly Notices of the Royal Astronomical Society*, 11 Jan. 2025, https://doi.org/10.1093/mnras/staf046.